Yiwei Chen

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Education

Michigan State University *East Lansing, MI, USA*

- **Ph.D. in Computer Science** Aug. 2024 -
  Advisor: Prof. Sijia Liu

Xi'an Jiaotong University *Xi'an, Shaanxi, China*

- **M.S. in Computer Science and Technology.** Ranking: 7/173 Sept. 2021 - Jun. 2024
- **B.Eng. in Automation.** Sept. 2017 - Jun. 2021
  Honor Class in Qian Xuesen Honors College. Ranking: 3/24
- **Special Class for Gifted Young.** Sept. 2015 - Jun. 2017
  A honor program for nationwide selected gifted young students.

Research Interests

- Reasoning, Post-training, Agentic System, LLMs, MLLMs
- AI Safety, Alignment, Trustworthy Algorithms, Interpretability, Machine Unlearning

Publications & Manuscripts

(* means Equal Contribution)

[1] **Y. Chen***, Y. Yao*, Y. Zhang, B. Shen, G. Liu, S. Liu. "Safety Mirage: How Spurious Correlations Undermine VLM Safety Fine-Tuning and Can Be Mitigated by Machine Unlearning." **ICLR**, 2026. [Paper]

[2] **Y. Chen***, S. Pal*, Y. Zhang, Q. Qu, S. Liu. "Unlearning Isn't Invisible: Detecting Unlearning Traces in LLMs from Model Outputs." **ICLR**, 2026. [Paper]

[3] **Y. Chen**, B. Shang, S. Liu. "Who Built This Model? Tracing LLM Lineage via Spectral Fingerprints in Weight Space." In submission, 2026. [Paper]

[4] **Y. Chen**, L. Li, K. Cheung, V. Parla, G. Sundaram. "Data-Centric Benchmarking of Exploit Generation in LLMs: Understanding the Impact of Fine-Tuning." [Paper]

[5] B. Shang*, **Y. Chen***, Y. Zhang, B. Shen, S. Liu. "Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning." In submission, 2026. [Paper]

[6] Z. Zhang*, **Y. Chen***, W. Zhang, C. Yan, Q. Zheng, Q. Wang, W. Chen. "Tile Classification Based Viewport Prediction with Multi-modal Fusion Transformer." **ACM-MM**, 2023. [Paper]

[7] Y. Zhang, C. Wang, **Y. Chen**, C. Fan, J. Jia, S. Liu. The Fragile Truth of Saliency: Improving LLM Input Attribution via Attention Bias Optimization. **NeurIPS Spotlight**, 2025. [Paper]

[8] H. Reisizadeh, J. Ruan, **Y. Chen**, S. Pal, S. Liu, M. Hong. Leak@k: Unlearning Does Not Make LLMs Forget Under Probabilistic Decoding. **ICML**, 2026. [Paper]

Internships

Cisco, Research Intern May 2025 - Oct. 2025
Research Project: Large Language Models for CVE Exploit Generation in Cybersecurity [4]

- Designed the first data-centric benchmark for CVE-conditioned exploit generation, including a 6-level context hierarchy, an 8-criterion evaluation framework, and large-scale benchmarking of 17 LLMs.
- Built a high-quality dataset via multi-stage refinement and fine-tuned Qwen3-8B with reasoning-aware

fine-tune, achieving +42.5% improvement and competitive performance with frontier LLMs.

Research Experience

OPTML, Michigan State University *East Lansing, US*
Research Assistant. Advisor: Prof. Sijia Liu Aug. 2024 -

- *MLLM Reasoning*: Existing improvements in mathematical reasoning for MLLMs are largely transferred from LLMs, with insufficient utilization of visual information during training. Propose visual-biased GRPO training that enhances visual grounding and improves reasoning performance.
- *LLM Agentic System*: Proposed a framework for multi-agent test-time training, decoupling inter-agent communication from fast-weight updates, and introducing local reasoning objectives to enable adaptive state refinement during inference.
- *MLLM Safety* [1]: Conventional safety fine-tuning of MLLMs suffers from a "safety mirage" caused by training bias, leading to spurious correlations and over-rejections under one-word attacks. Unlearning algorithms effectively remove harmful content and mitigate these issues.
- *LLM Unlearning* [2]: Investigated how machine unlearning leaves persistent, detectable "fingerprints" in both outputs and internal activations. Designed classifiers could identifying unlearned models, revealing new risks of reverse-engineering forgotten information.
- *LLM Lineage* [3]: Proposed a geometric fingerprinting framework to trace LLM lineage in weight space, combining spectral energy for coarse-grained discrimination and subspace alignment for fine-grained analysis, enabling data-free model provenance identification
- *Backdoor Unlearning* [5]: Demonstrated that LLM unlearning can be compromised through attention-sink-guided backdoor unlearning, where triggers placed at attention sinks enable models to recover forgotten knowledge while maintaining normal behavior without triggers.

Awards & Honors

SCHOLARSHIPS

- Michigan State University Travel Award — Oct. 2025
- Outstanding Master Graduate, Xi'an Jiaotong University (Top 5% of all graduates) — Jun. 2024
- Outstanding Bachelor Graduate, Xi'an Jiaotong University (Top 5% of all graduates) — Jun. 2021
- Outstanding Freshman Scholarship, Xi'an Jiaotong University (Top 10%) — Nov. 2021
- First Prize Scholarship, Xi'an Jiaotong University (Top 10%) — Nov. 2016-2019, 2022-2023

COMPETITIONS

- Meritorious Winner in Mathematical Contest in Modelling (top 8% in 25370 teams) — Mar. 2019
- The First Prize in Shaanxi of China Undergraduate Mathematical Contest in Modeling — Oct. 2018

Academic Service

- **Conference Reviewer:** ICLR, ICASSP, NeurIPS, ACM-MM, IJCNN
- **Journal Reviewer:** Journal of Machine Learning Research (JMLR)

Language & Skills

Language	Mandarin (Native), English (Professional)
Programming	Python, C/C++, Bash, LaTeX, MATLAB, HTML/CSS
Model Frameworks	LLMs (e.g., Llama, Qwen), MLLMs (e.g., LLaVA, Qwen-VL)
Libraries / Softwares	Pytorch, Tensorflow, Deepspeed, Huggingface, OpenCV
Developer Tools	Git, Docker, Vim, VSCode, Cursor, Claude Code, PyCharm